Whisker Scope

Manual. Free and open source by Studio 87. Support: ko-fi.com/studio87

Intro

Whisker Scope shows what your sound is doing: its waveform, its frequencies, how wide it is, and how close it is to a sound you want to match. Everything you see is measured from a copy of the audio. The sound goes through untouched unless you use Input gain, turn on the matching EQ on the Shapes page, use Output gain, or press Panic.

It is free and open source, made by Studio 87. It collects nothing and works fully offline.

Quick start

- Put Whisker Scope on the track or the master you want to look at. Play something.
- Pick a page at the top: Scope, Spectrum, Stereo field or Shapes.
- Hover the Spectrum to read the frequency, the note and the level under the mouse.
- Input gain changes the level going into the plugin: the sound, the displays and the meters all follow it. SC gain only changes what you see of the sidechain.
- Hover any control for one second to read what it does. Settings can turn those bubbles off.

Scope

The waveform of the sound. Channels picks L / R, left, right, mono or mid / side. Sync keeps the picture steady: Free shows the newest samples, Trigger starts each sweep at a rising zero crossing, Pitch locks to the detected pitch and shows two periods, Beat shows one beat of the host tempo. Visible time sets how many milliseconds you see. Freeze holds the picture.

Spectrum

Which frequencies are in the sound. FFT size trades sharpness (bigger, good for bass) against speed (smaller). Speed sets how fast the picture falls. Range sets how many dB are shown. Smoothing averages neighbouring frequencies. Slope tilts the picture by dB per octave around 1 kHz (3 dB makes pink noise look flat). Linear switches to a linear frequency scale, Fill colours under the line, Peak hold keeps the highest value of every frequency until Peak reset. The highest peak is labelled with its level, frequency and note.

Key readout

The box in the top right corner of the Spectrum guesses the key of what you play: the key name, the Camelot code (8A is A minor, 8B is C major) and a confidence value. It listens to about 6 seconds of the sound between 100 Hz and 2 kHz and compares the notes it hears with typical major and minor profiles (Krumhansl-Kessler). It needs a sustained tonal sound: chords, pads, a melody over some bars. It shows "Listening" at first and "No clear key" for noise, drums only or sweeps. A single chord is ambiguous with its relative key (C major and A minor share the same notes), so the box then shows "or" and the second guess; it is a hint, not a verdict. Key hold freezes the readout so it stays while you play something else. Key reset forgets what it heard and starts again.

Stereo field

The goniometer shows where the sound sits between left and right. Straight up is mono, sideways is wide, a diagonal means one side only. The correlation meter goes from -1 (antiphase, cancels in mono) through 0 (stereo) to +1 (mono). Width is the share of side energy. Mono loss is how much quieter the sound gets when left and right are added. The verdict (Mono safe, Check mono, Phase problem) follows simple rules of thumb: it is a hint, not a standard.

Shapes

Shapes compares the average spectrum of a reference (a song, a lead you love) with your own sound, on 60 bands of 1/6 octave from 20 Hz, Mid and Side kept apart.

- Press Capture reference, play the reference, press the button again. Only moments with signal are averaged.
- Press Capture mine and do the same with your sound. Without it, the live spectrum is compared.
- The bars at the bottom are the EQ that moves your sound toward the reference (pink cuts, teal boosts). Both shapes are first levelled to their own average, so loudness does not matter, only the balance of frequencies.
- Strength scales the curve. Max move limits the largest boost or cut.
- EQ on turns the matching Mid/Side EQ on in the audio path, so you can hear it. It is 30 bells per channel and every control is automatable. Auto level keeps your loudness or matches the loudness stored with the reference. Input and Output set the level around the EQ. A safety limiter sits at the very end.
- Load file analyses an audio file on your computer and uses its average Mid and Side shape as the reference, with a progress bar and a Cancel button. It runs only on your computer and nothing is uploaded. WAV, AIFF, FLAC and OGG always work; MP3 and others only where your system can decode them (macOS and Windows usually can, Linux builds do not read MP3). Files shorter than 2 seconds or longer than 60 minutes are refused, and silent parts are skipped.
- Save as preset keeps the current reference (captured or from a file) under a name in your library, in the Studio 87 / Whisker Scope folder on this computer. The drop-down "Or pick a reference preset" lists the starting points and your own. Manage renames, deletes, exports (as text) and imports presets. Bad text is refused.
- The starting points are made by formula and are not measured from any song.
- Copy EQ points copies the bells (frequency, gain, Q) for Mid and Side, to type into any other EQ.

Sidechain layer (masking)

Route a second track into the plugin's sidechain input in your host (layer B). It is drawn in yellow on the Spectrum, Scope and Stereo field and is never added to the output. On the Spectrum, Masking marks in red the frequency ranges where your track and the sidechain track are within the chosen distance (3, 6 or 10 dB) of each other, which is where they may mask each other. SC gain only changes what you see.

Window

Drag any edge or corner to resize; the shape stays the same (25 % to 250 %). Double-click a wooden side for the default size. The size is remembered.

Levels

On the right: level bars for left and right (IN and OUT on the Shapes page) with peak and RMS readouts, crest factor, a held peak (Peak reset clears it) and the CLIP light. Metering and RMS length choose how they react. Below them are correlation, width, balance, mono loss and the Mono safe verdict. These are monitors, not certified meters.

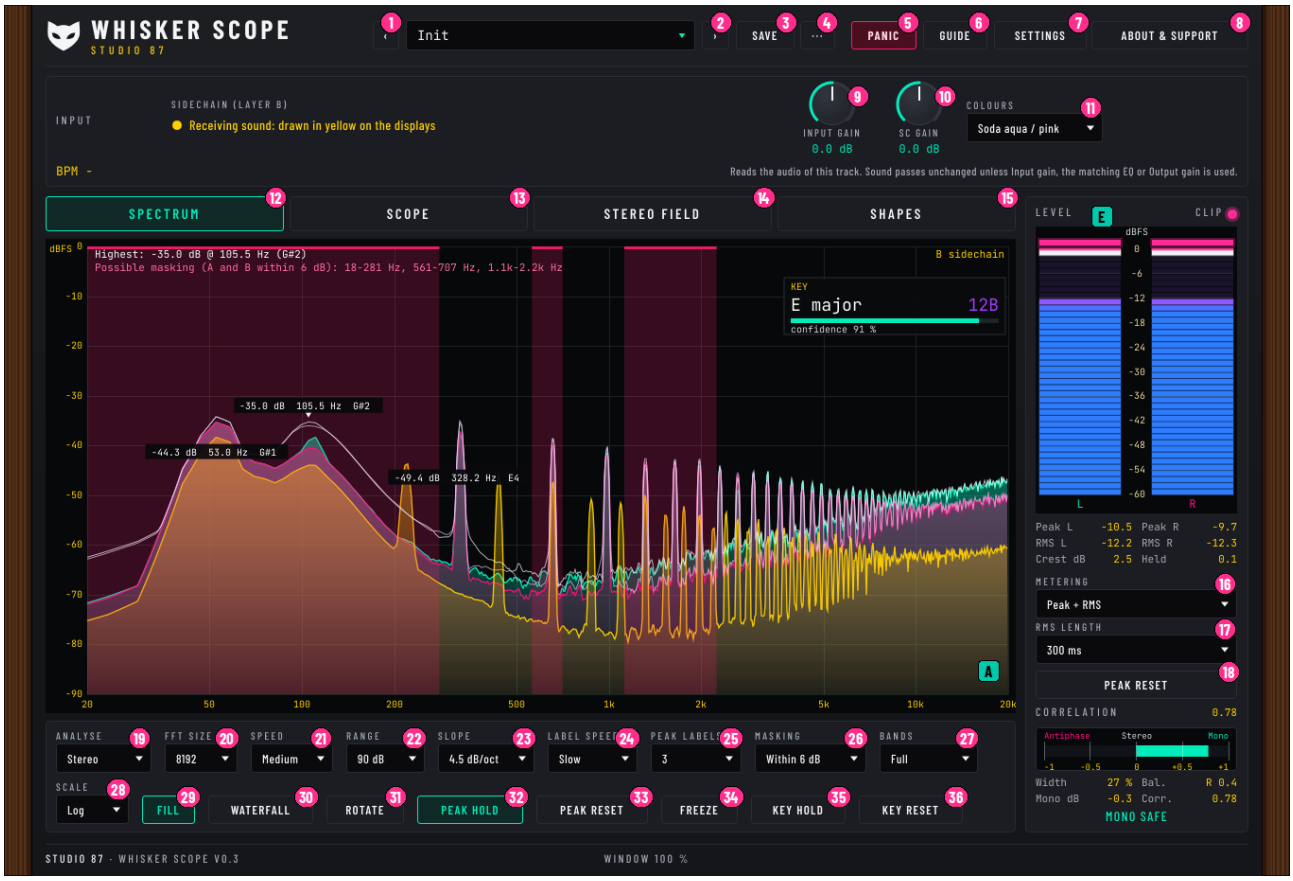
Recommendations

- Match shapes with the same kind of material: a lead against a lead, a full mix against a full mix.
- Treat the EQ curve as a hint. Move a little, then listen.

- Use Mono safe as a quick check before you export, and listen in mono once.
- Capture for a few seconds of the busiest part of the song, not the intro.

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The pictures are screenshots of the real plugin window (rendered offscreen from the plugin code with a made-up test signal), not a mockup.



Signal flow

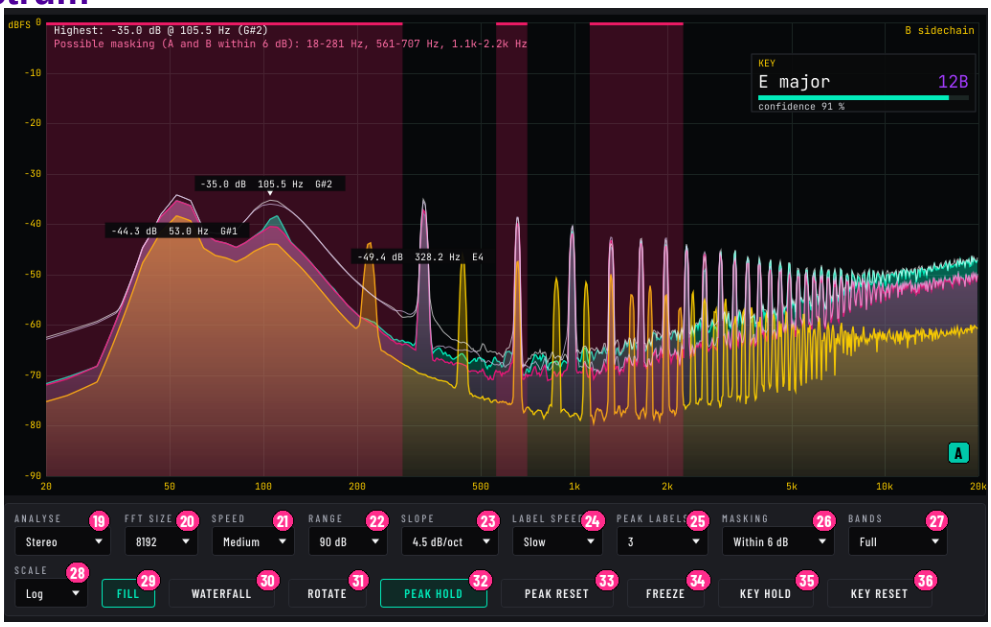
Input (your track) > Input gain > Mid/Side match EQ (Shapes page, off until you switch EQ on) > Auto level and Output gain > safety limiter > output. The displays and meters only read the audio; with everything at default the sound passes through bit for bit. The sidechain input (layer B) is read only for the yellow traces and the masking view; it is never added to the output.

Header and tabs

1	**Previous preset**	Previous preset.
2	**Next preset**	Next preset.
3	**Save preset**	Save the current settings as your own preset (stays on this device).
4	**More**	Rename, delete, export or import presets.
5	**Panic**	PANIC: mutes the output at once, cancels any capture and turns the matching EQ off. Press again to unmute. Shortcut: P.
6	**Guide**	Quick guide: what each view and control does.
7	**Settings**	Info bubbles, colours, power saving, meter warning level, window size.
8	**About & Support**	About Studio 87, privacy, installing, and ways to support the project.
9	**Input gain**	Boosts or cuts the level going into the plugin. The displays, the meters and the matching EQ all follow it. Double-click resets.

10	**SC gain**	Level of the sidechain (layer B) signal that the displays see. It is never played. Double-click resets.
11	**Colours**	Colour scheme for all displays.
12	**Spectrum**	Spectrum analyser with optional waterfall. Shortcut: 1.
13	**Scope**	Waveform scope. Shortcut: 2.
14	**Stereo field**	Stereo field: goniometer, phase correlation and mono check. Shortcut: 3.
15	**Shapes**	Freeze the spectrum shape of a song or lead, save it, and see the EQ that moves your sound toward it. Shortcut: 4.

A. Spectrum



Display A: Spectrum. Horizontal axis: frequency, 20 Hz to 20 kHz (log or linear with Scale). Vertical axis: level in dBFS (range 60, 90 or 120 dB, default 90 dB). Teal and pink are the left and right channels, white is the peak hold line, yellow is the sidechain (layer B). Red bands show where A and B are within the Masking distance of each other, which is where they may mask each other. The labels name the loudest peaks with note names. The KEY box in the top right guesses the key (with the Camelot code and a confidence value) of sustained tonal sound.

19	**Analyse**	Which signal the analyser looks at. Stereo (left and right) and Mid / Side draw two curves. (shown: Stereo)
20	**FFT size**	Bigger sizes separate bass notes better but react slower. (shown: 8192)
21	**Speed**	How quickly the curve follows the sound. (shown: Medium)
22	**Range**	How many dB below 0 dBFS the display shows. (shown: 90 dB)
23	**Slope**	Tilts the curve up toward the highs so a typical mix looks flat. Pivot is 1 kHz. 4.5 dB/oct is the default, the same value other analysers use. (shown: 4.5 dB/oct)
24	**Label speed**	How often the peak labels refresh. Slower is easier to read. (shown: Slow)
25	**Peak labels**	Shows the peak level and the pitch (Hz and note) right on the scope. 0 hides them. (shown: 3)
26	**Masking**	With a sidechain (layer B) playing: marks the frequency ranges where A and B are both loud and within this many dB of each other, a sign that they may mask each other. Rule of thumb, check by ear. (shown: Within 6 dB)

27	**Bands**	Full shows every detail. Octave bands show how an equaliser would see it. (shown: Full)
28	**Scale**	Log spaces the octaves evenly. Linear spaces the Hz evenly. (shown: Log)
29	**Fill**	Switches between a filled spectrum and a thin line.
30	**Waterfall**	Draws the history of the spectrum as colours scrolling away.
31	**Rotate**	Turns the waterfall sideways: time runs left, frequency runs up.
32	**Peak hold**	Keeps a thin line at the loudest value each frequency reached.
33	**Peak reset**	Clears the held peaks and the clip light. Shortcut: R.
34	**Freeze**	Holds the spectrum picture. Shortcut: F.
35	**Key hold**	Freezes the key readout in the top right corner of the display, so it stays while you play something else.
36	**Key reset**	Forgets the notes the key detector has heard so far and starts listening again. Use it when the song changes key or you load a different piece.

B. Scope

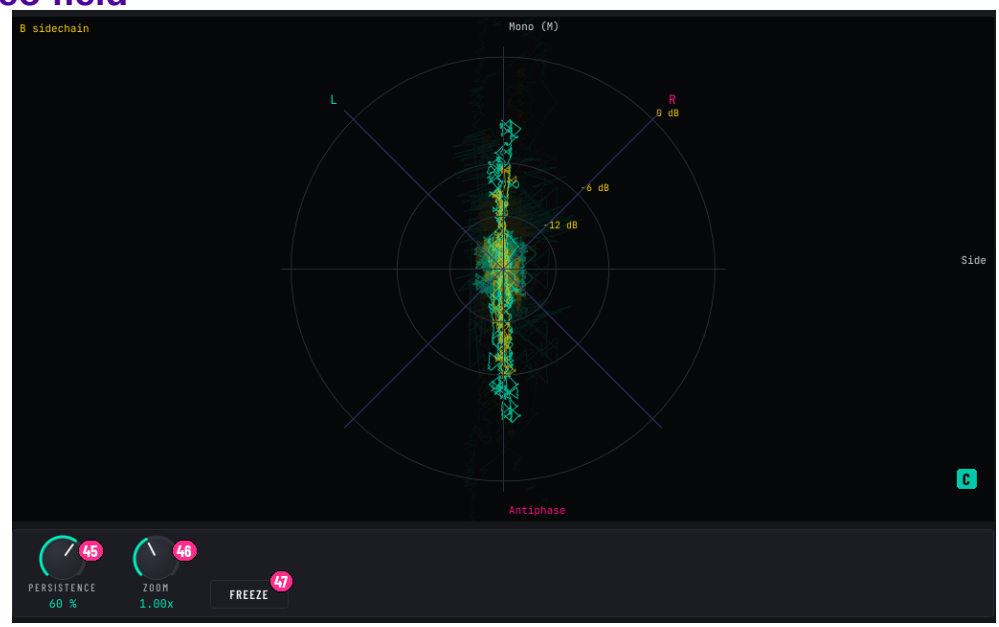


Display B: Scope. Horizontal axis: time (the Time knob, 60 ms in the picture). Vertical axis: level in dB around the centre line, so quiet sound is still visible. Teal is left, pink is right, yellow is the sidechain. Sync modes keep the picture still: Free, Trigger, Frequency, Pitch and Beat (follows the host tempo). Transients marks hits with vertical lines.

39	**Channels**	Which channels are drawn. Mid is (L+R)/2, Side is (L-R)/2. (shown: Stereo)
40	**Sync**	Free: always shows the latest audio. Trigger: locks on a rising edge so repeating waves hold still. Frequency: fixed Hz you set. Pitch: follows the detected pitch. Beat: window follows the host tempo. (shown: Free)
37	**Time**	How much time fits on the screen (1 ms to 400 ms). Double-click resets.
38	**Trig level**	Level the Trigger mode waits for before starting the sweep. Double-click resets.
41	**Label speed**	How often the peak labels refresh. Slower is easier to read. (shown: Slow)
42	**Peak label**	Shows the peak level and the pitch (Hz and note) right on the scope. 0 hides them. (shown: 3)

43	**Transients**	Starts the window on the latest sudden loud event, handy for kicks and plucks.
44	**Freeze**	Holds the scope picture. Meters keep moving. Shortcut: F.

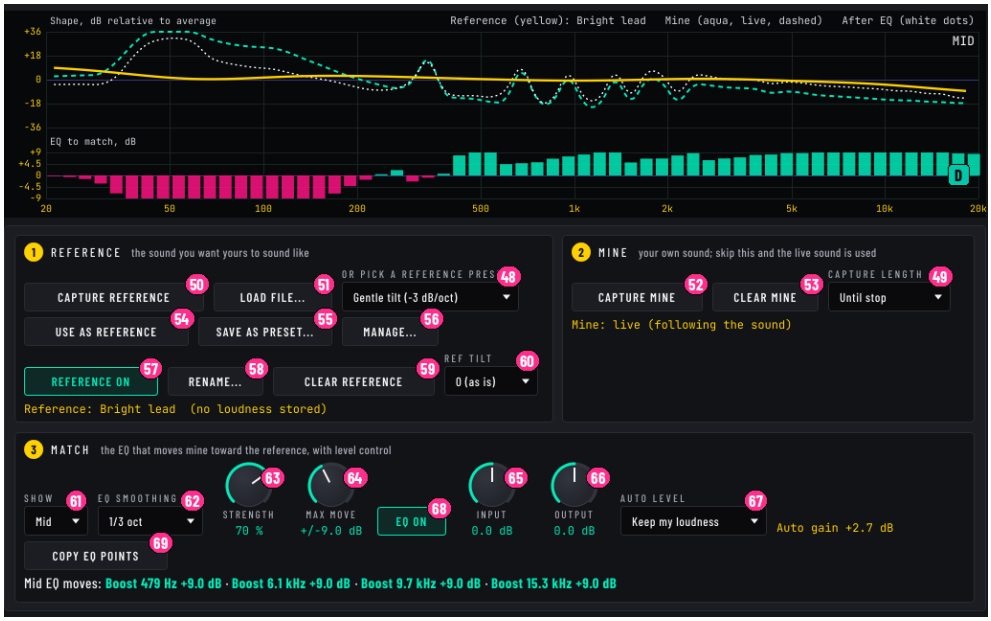
C. Stereo field



Display C: Stereo field (vectorscope). Vertical axis: Mono (M), the sound both channels share. Horizontal axis: Side. Left and Right are the diagonals; Antiphase is the bottom. A narrow vertical shape is mono, a wide cloud is stereo, anything below the centre line points to phase trouble. The rings mark 0, -6 and -12 dB. Persistence controls how long the trail stays, Zoom magnifies quiet sound.

45	**Persistence**	How long the dots leave a trail. More trail shows the shape of the sound over time. Double-click resets.
46	**Zoom**	Magnifies the picture so quiet material becomes visible. Double-click resets.
47	**Freeze**	Holds the picture. Shortcut: F.

D. Shapes

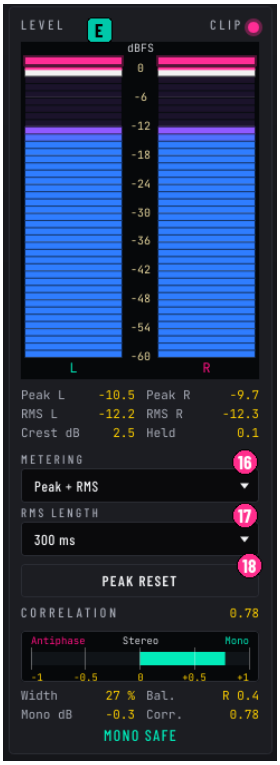


Display D: Shapes. Top graph: average spectrum shape in dB relative to its own average (horizontal 20 Hz to 20 kHz). Yellow is the reference, dashed aqua is yours, white dots are yours after the EQ. Bottom graph: the EQ that moves yours toward the reference, in dB (pink cuts, teal boosts), limited by Max move. The label in the corner shows Mid or Side. Reference: Load file analyses an audio file on this computer, the drop-down lists reference presets, Save as preset keeps one.

50	**Capture reference**	Play the song or lead you want to sound like, press this, let it play, then press again. Its average spectrum shape and loudness are frozen as the reference.
51	**Load reference file**	Analyses an audio file on this computer and uses its average Mid and Side shape as the reference. Works offline, nothing is uploaded. .wav .bwf .aiff .aif .flac .ogg files can be read on this system.
48	**Reference presets**	Starting-point shapes made by formula (never measured from a song) plus the ones you saved. (shown: Gentle tilt (-3 dB/oct))
54	**Use as reference**	Loads the chosen shape as the target.
55	**Save reference as preset**	Keeps the current reference (shape and loudness), captured or loaded from a file, as a named preset in your library on this device.
56	**Manage reference presets**	Rename, delete, export or import reference presets.
57	**Reference on**	Switches the reference off without losing it: the comparison and the matching EQ stop, the shape stays stored. Turn it back on any time.
58	**Rename reference**	Gives the current reference a name of your choice.
59	**Clear reference**	Removes the reference. Mine, the EQ settings and everything else stay. Save it first (Save as) if you want to keep it.
60	**Ref tilt**	Edits the reference: tilts its brightness around 1 kHz, for example to make a dark reference a bit brighter. 0 leaves it as it is. (shown: 0 (as is))
52	**Capture mine**	Play your own sound, press this, let it play, then press again. Its shape is frozen so you can compare while you move on.
53	**Clear mine**	Forget the captured shape of your sound and compare the live sound again.
49	**Capture length**	How long a capture averages the sound. (shown: Until stop)

61	**Show**	Which half of the sound the graph shows. Mid is the centre (kick, bass, lead), Side is what is wide. Both halves are always matched. (shown: Mid)
62	**EQ smoothing**	Wider smoothing gives gentler, more musical EQ moves. (shown: 1/3 oct)
63	**Strength**	How much of the difference to correct. 100 % tries to copy the reference exactly, 50 % goes halfway. Double-click resets.
64	**Max move**	No EQ move is allowed to be bigger than this. Double-click resets.
68	**EQ on**	Turns the matching EQ on or off so you can A/B. With Auto level on, the loudness stays the same, so you hear only the tone change. The EQ is 30 bells for Mid and 30 for Side, a third of an octave apart.
65	**Input**	Level into the plugin (same knob as the Input gain at the top). Double-click resets.
66	**Output**	Level out of the plugin, after the EQ and the auto level. Double-click resets.
67	**Auto level**	Off: only the Output knob sets the level. Keep my loudness: the EQ never makes the sound louder or quieter than what comes in, so you judge tone, not volume. Match reference: output loudness follows the loudness captured with the reference (needs a captured or saved reference that has loudness). (shown: Keep my loudness)
69	**Copy EQ points**	Copies the matching EQ as a list of bells (frequency, gain, Q) for Mid and Side, to type into any EQ.

E. Meters



Display E: Level meters for the left and right channel in dBFS (peak with hold line, RMS in the body), clip light, and the readouts for peak, RMS, crest factor and held peak. Below: the correlation meter (-1 antiphase, +1 mono), stereo width, balance, mono loss in dB and a verdict (MONO SAFE and so on).

16	**Metering**	Which level the bars show. Peak is the loudest moment, RMS is the average level, Loudness (LUFS) is how loud it sounds (momentary, short-term and integrated, K-weighted as in streaming and broadcast meters). (shown: Peak + RMS)
17	**RMS length**	Averaging time of the RMS meter. Longer is calmer and closer to perceived loudness. (shown: 300 ms)

18	**Peak reset**	Clears held peaks, the clip light and the crest figure. Shortcut: R.

Under the hood

• Everything you see is read from a copy of the audio. The audio path itself is untouched unless you turn on the matching EQ, use Output gain, or press Panic.

• The matching EQ is a Mid/Side EQ made of 30 peaking bells per channel (Q 2.5). The Shapes analysis runs about 20 times per second on the interface thread, never in the audio callback. Strength scales the EQ, Max move limits each bell, EQ on bypasses it.

• Auto level keeps your loudness (the EQ is not allowed to make the sound louder or quieter) or matches the loudness captured with the reference. A safety limiter at the end stops a wrong setting from sending a huge level to your speakers. Panic mutes the output.

• The spectrum uses an FFT of 2048 to 32768 points with a Blackman window; larger sizes resolve low notes better and react slower. The sidechain is the second input bus; if your host does not route anything there, the yellow traces stay off.

• Every knob, menu and switch is an automatable plugin parameter. Presets and saved shapes are plain files in the Studio 87 / Whisker Scope folder of your user data; nothing is sent anywhere.

• The window keeps its shape and scales from 25 % to 250 %. Drag any edge or corner, or double-click a wooden side for the default size.